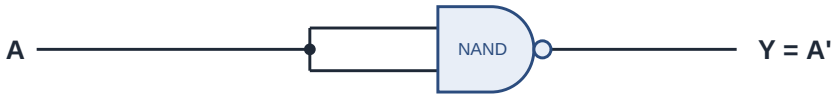
NAND Universal Gate

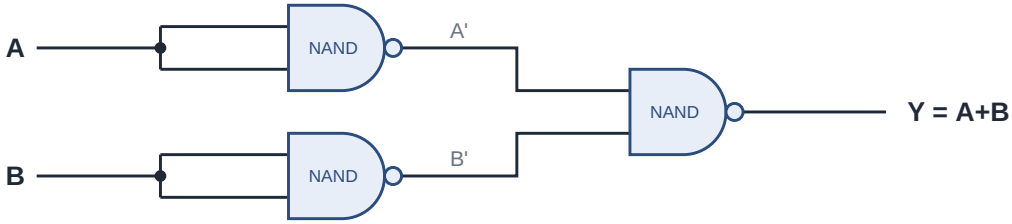
NOT, AND, OR built using only NAND gates

NOT = 1 NAND (both inputs tied)



AND = 2 NAND

OR = 3 NAND



Boolean proof (step by step)

NOT : Y = (A.A)' = A'

AND : N1 = (A.B)' -> Y = (N1.N1)' = ((A.B)')' = A.B

OR : A' = (A.A)', B' = (B.B)' -> Y = (A'.B')' = A + B [De Morgan]

Yaad rakhne ka shortcut

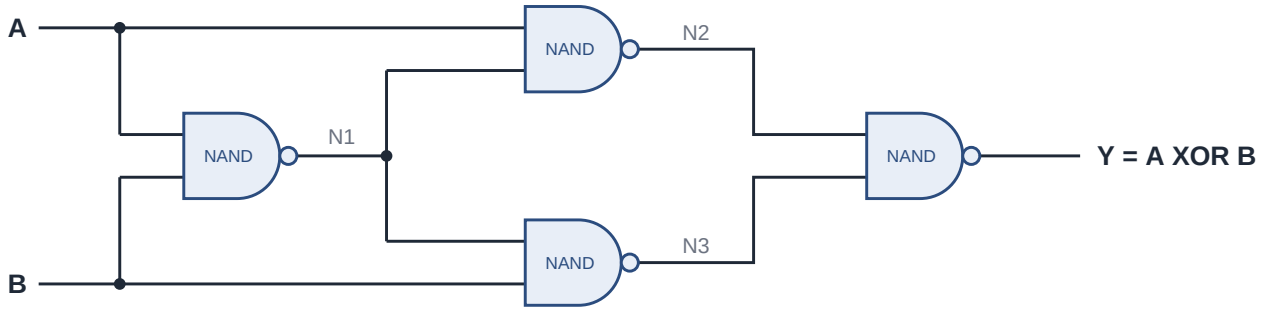
AND = NAND + NOT OR = (NOT A) NAND (NOT B)

NOR = OR + NOT XNOR = XOR + NOT

XOR and XNOR using NAND

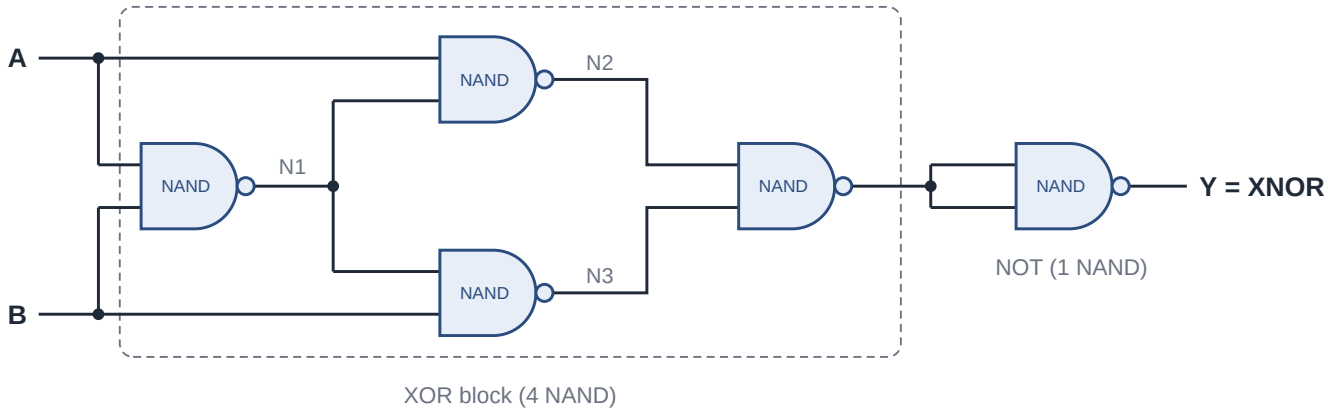
XOR = 4 NAND, XNOR = 5 NAND

XOR = 4 NAND



N1 = (A.B)' N2 = (A.N1)' N3 = (B.N1)' Y = (N2.N3)' = A'B + AB'

XNOR = 5 NAND (XOR + NOT)

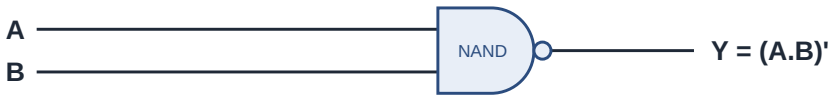


Y = (XOR)' = AB + A'B' [A aur B same ho to output 1]

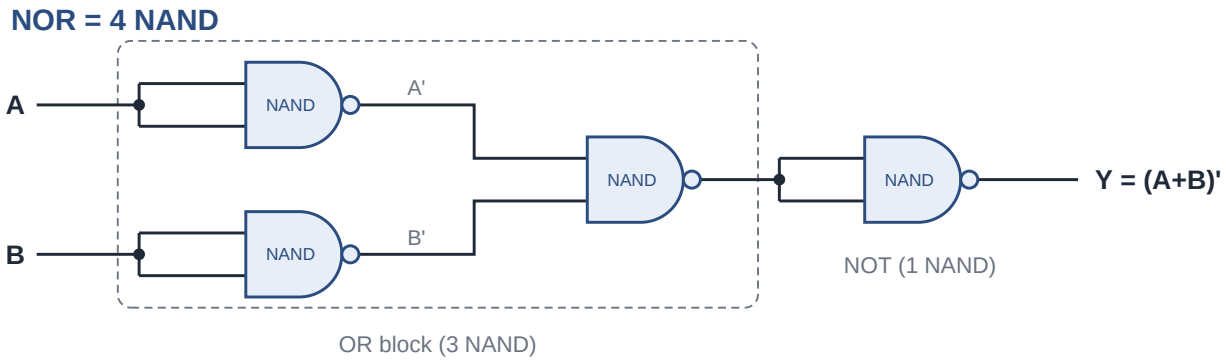
NAND and NOR using NAND

NOR = 4 NAND

NAND = basic gate (1 NAND)



NOR = 4 NAND



NOR : OR = (A'.B')' = A+B -> Y = (OR.OR)' = (A+B)'

Summary table (exam ke liye)

Gate	NAND count	Idea
NOT	1	Dono inputs tied: (A.A)' = A'
AND	2	NAND ke baad NOT
OR	3	(A'.B')' = A+B (NOT inputs, phir NAND)
NAND	1	Khud basic gate
NOR	4	OR ke baad NOT
XOR	4	N1=(AB)', N2=(A.N1)', N3=(B.N1)', Y=(N2.N3)'
XNOR	5	XOR ke baad NOT